

Adaptive Hybrid Retrieval for Multi-Hop Question Answering: Dynamic Routing Between Dense and Graph-Guided Retrieval, with Vectorless Retrieval as an Exploratory Baseline

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Abstract—Retrieval-augmented generation (RAG) has emerged as an effective paradigm for grounding large language models in external evidence, yet many existing systems still rely on a fixed retrieval strategy. This design becomes restrictive in multi-hop question answering, where answering a single query may require semantic matching, relational expansion, and navigation across distributed evidence. Dense retrieval is efficient and widely used, but it may fail to preserve multi-step evidence chains. Graph-guided retrieval captures relations among entities and passages more explicitly, although its benefits are not uniform across all query types and it introduces additional indexing overhead. Structure-aware retrieval methods further suggest that hierarchical document organization can improve evidence access in long-context settings. Motivated by these complementary strengths and limitations, this paper investigates an adaptive hybrid retrieval framework for multi-hop question answering. The proposed system dynamically routes queries between dense retrieval, graph-guided retrieval, and a vectorless structural branch treated as an exploratory baseline. Four retrieval configurations are implemented and evaluated on a HotpotQA subset: Dense RAG, Graph RAG, Hybrid RAG, and PageIndex RAG. Hybrid RAG achieves the highest Exact Match of 0.106 and Token F1 of 0.178 while also exhibiting substantially reduced latency relative to graph-guided retrieval. The study further identifies a gate miscalibration caused by the negative score distribution of FAISS IndexFlatL2, analyzes the MRR paradox in which graph expansion reduces ranking precision while improving answer quality, and discusses dataset leakage introduced by same-context edge construction. These findings highlight concrete directions for improving adaptive retrieval in multi-hop settings.

Index Terms—retrieval-augmented generation, multi-hop question answering, adaptive routing, graph-guided retrieval, dense retrieval

I. INTRODUCTION

Retrieval-augmented generation (RAG) improves knowledge-intensive question answering by combining a language model with an external retriever rather than relying solely on parametric memory. The retrieval module is therefore central to system quality, particularly when the answer cannot be inferred from a single passage but must instead be assembled from several dependent pieces of evidence [1], [2]. This challenge is especially visible in

multi-hop question answering, where relevant facts are often distributed across documents and must be connected through intermediate reasoning steps [3], [4].

A major limitation of many current RAG pipelines is their dependence on a single retrieval paradigm. Dense retrieval remains the dominant choice because it is simple, scalable, and effective for semantic matching. However, multi-hop question answering frequently requires evidence gathering that is conditional and sequential rather than purely similarity-based [2]. MultiHop-RAG demonstrates that conventional retrieve-then-generate pipelines degrade when support must be drawn from multiple documents and combined through multi-step reasoning [3]. Even when relevant passages are retrieved, the presence of noisy or only partially relevant evidence can still reduce downstream answer quality [5].

Graph-guided retrieval provides a different retrieval bias by explicitly modeling relations among entities, chunks, or documents. HippoRAG shows that graph-structured memory can support multi-step retrieval through associative propagation, while KG²RAG starts from semantic seed retrieval and then expands evidence using knowledge graph relations to improve coherence and diversity [6], [7]. HybGRAG further demonstrates that some questions require both textual and relational evidence, which suggests that neither dense retrieval nor graph retrieval alone is universally sufficient [8]. Comparative analysis also indicates that dense RAG and GraphRAG exhibit complementary strengths rather than a single dominant behavior across tasks [9].

A third perspective emerges from structure-aware retrieval. RAPTOR improves retrieval over long documents by organizing information into a hierarchical summary tree rather than relying only on flat chunk matching [10]. Edge *et al.* likewise show that graph-based organization can support local and global evidence access for query-focused summarization [11]. Although vectorless retrieval remains less established in the peer-reviewed literature than dense and graph-guided methods, these structure-oriented approaches suggest that hierarchical document organization may provide an alternative retrieval

signal for complex or long-context queries.

The motivation of this paper is therefore to investigate whether retrieval itself should be adaptive. Rather than assuming that one retriever is optimal for all questions, the paper considers a hybrid framework in which dense retrieval addresses local semantic queries, graph-guided retrieval addresses relation-heavy multi-hop queries, and a vectorless structural branch is treated as an exploratory baseline for structure-sensitive questions. The objective is to establish the literature foundation for a routed retrieval design that dynamically selects among these paradigms for multi-hop question answering [1], [7], [8], [10].

II. RELATED WORK

The literature on multi-hop question answering consistently shows that one-shot retrieval is poorly matched to questions whose supporting facts must be discovered and combined across several reasoning steps. IRCoT addresses this mismatch by interleaving retrieval with chain-of-thought reasoning so that intermediate inference can guide subsequent evidence acquisition [2]. MultiHop-RAG extends this concern from system design to evaluation by constructing queries whose evidence is intentionally distributed across documents, thereby exposing weaknesses in standard RAG pipelines [3]. HotpotQA remains important in this setting because it provides supporting-fact supervision and therefore makes retrieval quality observable rather than purely implicit in the final answer [4]. Together, these works establish that multi-hop QA requires evidence composition rather than simple nearest-neighbor retrieval.

Graph-based retrieval methods address this issue by making relationships among evidence units explicit. HippoRAG models long-term knowledge through a graph-like memory structure that improves multi-step retrieval via concept associations [6]. KG²RAG similarly begins with semantic retrieval but then expands and organizes evidence using knowledge graph relations, improving coherence for downstream generation [7]. HybGRAG sharpens this argument by showing that hybrid questions over semi-structured knowledge bases often require both relational traversal and textual grounding, which neither dense retrieval nor graph retrieval can provide alone [8]. Han *et al.* reinforce this conclusion through systematic comparison, reporting that dense RAG remains competitive on detail-oriented tasks while GraphRAG-style approaches are more advantageous when broader relational aggregation is required [9]. The implication is not that graph-guided retrieval should replace dense retrieval, but that different query types benefit from different retrieval biases.

Structure-aware retrieval contributes a complementary line of evidence. RAPTOR demonstrates that hierarchical abstraction can improve retrieval over long documents by retrieving across multiple levels of a summary tree rather than only over flat chunks [10]. Edge *et al.* similarly show that graph-based corpus organization can support both local and global evidence access in query-focused retrieval settings [11]. These methods do not constitute a mature vectorless literature in

the same sense as dense or graph-guided retrieval, yet they strongly suggest that document structure can function as an independent retrieval signal. This observation motivates the exploratory vectorless branch in the present topic, even though the primary scholarly foundation remains dense, graph-guided, and adaptive retrieval research.

The adaptive dimension is most directly represented by Adaptive-RAG, which formulates retrieval selection as a query-complexity decision and routes questions to strategies of differing cost and sophistication [1]. The broader survey literature on agentic RAG further indicates that retrieval pipelines are increasingly being treated as controllable processes involving planning and dynamic strategy selection rather than static modules [12]. However, much of this work remains either architectural or survey-oriented. A narrower empirical gap remains at the intersection of multi-hop QA and hybrid retrieval: most existing studies optimize one retrieval paradigm at a time, while the accumulated evidence suggests that different questions favor dense, graph-guided, or structure-oriented retrieval for different reasons. This gap motivates the proposed direction of adaptive hybrid retrieval, where routing becomes the central mechanism for deciding which retrieval paradigm should be applied to a given multi-hop query.

III. PROBLEM IDENTIFICATION AND MOTIVATION

Multi-hop question answering presents a class of queries for which the answer cannot be obtained by retrieving a single relevant passage. Instead, answering requires assembling evidence from multiple documents, each of which may address only one step in a chain of reasoning. A question such as “Who directed the film in which the actor who played the villain in film X also appeared?” demands locating distinct facts from separate contexts and composing them correctly. Standard RAG pipelines that retrieve the top- k passages by semantic similarity and pass them to a reader are poorly suited to this structure, because the passages that are individually most similar to the question may not collectively provide the full evidence chain [3].

Dense retrieval, which encodes queries and documents into a shared embedding space and retrieves by nearest-neighbor search, is effective when the answer is contained in a single passage that closely matches the query semantically. In multi-hop settings, however, intermediate evidence passages may not be semantically close to the original query, because they address only one step in the reasoning chain rather than the final question. This creates a coverage gap: the retrieval model may consistently locate the first evidence hop while missing the second, leaving the language model without the information it requires to answer correctly.

Graph-guided retrieval addresses this limitation by explicitly encoding relationships among passages. Starting from semantically retrieved seed documents, a graph traversal can follow entity co-occurrence, hyperlink structure, or contextual overlap to locate passages that are not directly similar to the query but are connected to retrieved seeds through a reasoning path. This is particularly valuable for bridge questions, which

require finding an entity that connects two concepts, and for comparison questions, which require locating attributes of two separate entities and contrasting them. Despite this advantage, graph traversal introduces its own failure modes. Noisy edges, particularly those based on surface-level entity overlap, can propagate through the graph and retrieve passages that are thematically related but factually irrelevant. The BFS expansion further risks including passages that fill the evidence budget without contributing to the answer.

The motivation for an adaptive hybrid framework is that neither dense retrieval nor graph-guided retrieval is uniformly superior. Dense retrieval provides high-precision initial retrieval and favorable ranking behavior, as reflected in its MRR score, but it may miss evidence that is required only in later reasoning steps. Graph-guided retrieval can recover that evidence through structural expansion, but it does so at the cost of retrieval precision and substantially higher latency. A system that can assess the quality of its initial dense retrieval and selectively invoke graph expansion only when the dense evidence is insufficient would retain the precision of dense retrieval for simpler queries while extending its reach to more complex multi-hop cases.

The present study identifies three concrete problems that motivate this design. First, the recall ceiling under dense-only retrieval is low: no system can correctly answer a question if the supporting evidence is not included in the retrieved set. Second, graph expansion as implemented here does not reliably improve recall at the chain level, because BFS traversal starting from low-quality seeds may not reach the correct evidence within the hop budget. Third, the gating mechanism intended to route queries to graph expansion is subject to calibration failures when the score distribution of the retrieval index does not align with the threshold assumptions embedded in the gate condition. These problems are analyzed empirically in Section VII and motivate both the design choices described in Section IV and the directions for future work identified in Section IX.

IV. METHODOLOGY AND SYSTEM ARCHITECTURE

This section describes the four retrieval systems implemented in the study. All systems share a common document corpus derived from HotpotQA and a common reader interface, but differ in their retrieval mechanisms, evidence selection strategies, and language model backends. The overall architecture of the proposed Hybrid RAG system is illustrated in Fig. 1.

A. Dense RAG

The Dense RAG system encodes all corpus documents using the `multi-qa-MiniLM-L6-cos-v1` model, which produces 384-dimensional sentence embeddings optimized for semantic similarity. All document embeddings are stored in a FAISS `IndexFlatL2` index, which supports exact ℓ_2 nearest-neighbor search without compression or approximation. At query time, the question is encoded using the same model, and the top-5 passages by ℓ_2 distance are retrieved.

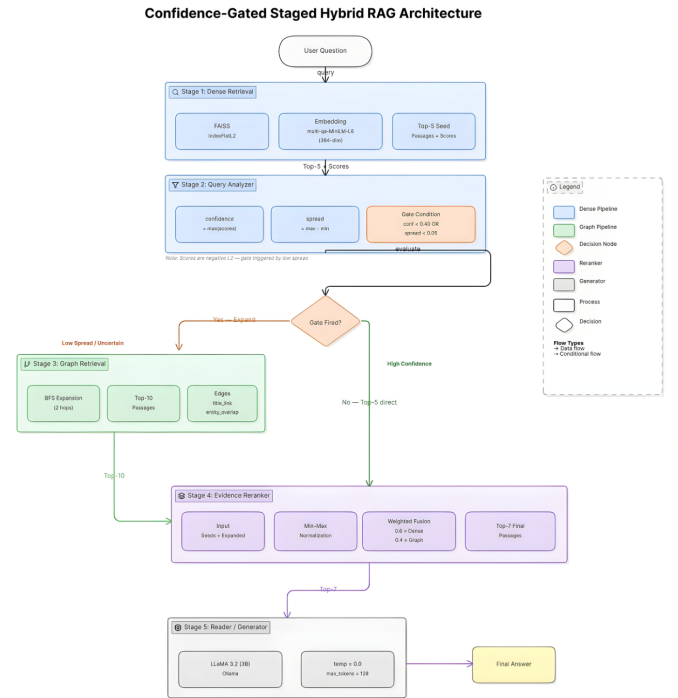


Fig. 1. System architecture of the Adaptive Hybrid RAG pipeline. The Query-Analyzer inspects dense retrieval confidence and score spread to determine whether BFS graph expansion is required. The evidence reranker combines dense and graph scores before final passage selection.

These passages are concatenated and provided as context to a locally hosted Llama 3.2 (3B) model served through Ollama. The model generates an answer in a single forward pass without iterative retrieval. Dense RAG serves as the primary baseline for evaluating the marginal contribution of graph expansion and adaptive routing.

An important property of FAISS `IndexFlatL2` is that it returns raw squared ℓ_2 distances rather than cosine similarity scores. As a consequence, all retrieval scores produced by this index lie in the range $[-1.41, 0]$ when computed as the negated ℓ_2 distance, with values closer to zero corresponding to higher similarity. This score distribution has direct implications for the gating mechanism in Hybrid RAG and is analyzed in detail in Section VII.

B. Graph RAG

The Graph RAG system extends dense retrieval with a breadth-first search expansion over a passage graph constructed offline. The graph is built over the HotpotQA corpus by creating directed edges between passages according to three criteria. A *title link* edge of weight 1.0 is added between any two passages whose titles appear in each other’s text, capturing explicit cross-document references. An *entity overlap* edge is added between passages that share named entities, with the weight capped to prevent high-degree entity nodes from dominating the graph structure. A *same-context* edge of weight

0.5 is added between passages that originate from the same HotpotQA context group, reflecting the dataset’s construction in which supporting facts are drawn from a limited number of documents per question.

At query time, the top-5 FAISS seeds are used as starting nodes for BFS expansion with a maximum hop depth of 2 and a total evidence budget of 10 passages. Retrieved passages are ranked by their graph traversal score and the top passages are concatenated as context for the reader. The Graph RAG system uses the Llama 3.2 (3B) model via Ollama as its reader, consistent with the Dense RAG configuration.

The same-context edge deserves particular scrutiny. Because it links passages that share a HotpotQA context identifier, it effectively encodes dataset structure into the graph topology. A system that exploits same-context edges may appear to perform multi-hop reasoning through graph traversal when it is in fact leveraging a shortcut derived from the evaluation dataset’s internal organization. This constitutes a form of retrieval leakage and is treated as a limitation in Section VIII.

C. Hybrid RAG

The Hybrid RAG system, which is the primary contribution of this study, implements a confidence-gated staged pipeline that combines dense retrieval and conditional graph expansion. The pipeline proceeds in four stages.

In the first stage, the query is embedded and the top-5 passages are retrieved from the FAISS index, identical to Dense RAG. In the second stage, a QueryAnalyzer module computes two signals from the FAISS score distribution: a *confidence* score defined as the maximum score among the top-5 results, and a *spread* value defined as the difference between the maximum and minimum scores. These signals serve as proxies for retrieval certainty: a high maximum score suggests that at least one passage is strongly relevant, while a low spread suggests that the score distribution is flat and no single passage is clearly more relevant than the others.

In the third stage, the gate condition is evaluated: if $\text{confidence} < 0.40$ or $\text{spread} < 0.05$, BFS graph expansion is triggered. When activated, the expansion follows the same graph structure as Graph RAG but excludes same-context edges, removing the leakage source. In the fourth stage, an evidence reranker assigns each passage a final score as a weighted combination of its dense score and graph score, specifically $\text{final_score} = 0.6 \times \text{dense} + 0.4 \times \text{graph}$. The top-7 passages by final score are selected and passed as context to the Llama 3.2 (3B) model via Ollama operating at temperature 0 to maximize answer determinism.

A critical finding regarding this gate is that the confidence threshold is never triggered in practice. Because FAISS IndexFlatL2 returns negated ℓ_2 distances in the range $[-1.41, 0]$, the maximum score among retrieved passages is always below zero, which is always below the threshold of 0.40. The gate therefore operates exclusively through the spread condition. This miscalibration means that the Hybrid system behaves as spread-gated rather than confidence-gated,

and the empirical results must be interpreted accordingly. This finding is discussed further in Section VII.

D. PageIndex RAG

The PageIndex RAG system is an exploratory baseline that performs retrieval without embeddings. It constructs an inverted keyword index over document titles and retrieves the top-10 candidate documents by keyword match to the query terms. These candidates are submitted to an external PageIndex API, and the system selects the answer from the API response based on citation density, defined as the frequency with which passages are referenced in the returned content. This system was designed to investigate whether structural and citation-based evidence alone could support question answering without any dense vector representation.

Due to access constraints encountered during the evaluation period, the PageIndex API was unavailable for systematic testing. PageIndex RAG is therefore treated as a conceptual baseline only: its design is described for completeness and to document the rationale for keyword-based retrieval, but no quantitative results are reported for this system.

V. DATASET, TOOLS, AND APIS USED

A. Dataset

All experiments use HotpotQA [4], a widely adopted multi-hop question answering benchmark constructed from Wikipedia articles. HotpotQA provides two types of annotations that are particularly valuable for this study: gold answers and supporting fact labels. The supporting fact labels identify the specific sentences from which a correct answer can be derived, enabling evaluation of retrieval quality at the evidence level in addition to final answer quality. Two question types are present in the dataset: bridge questions, which require locating an intermediate entity to connect two concepts, and comparison questions, which require retrieving attributes of two entities and contrasting them. The evaluation reported in this paper uses a randomly sampled subset of 500 questions from the HotpotQA development set in order to keep total inference time tractable under the local CPU inference constraint imposed by the Ollama-based reader.

B. Tools and Frameworks

The vector index for dense retrieval is implemented using FAISS [13], specifically the IndexFlatL2 configuration, which performs exact brute-force nearest-neighbor search and avoids approximation artifacts that could confound retrieval quality measurement. Document encoding uses the multi-qa-MiniLM-L6-cos-v1 model from the Sentence Transformers library, selected for its balance between embedding quality and inference speed on CPU hardware. Named entity recognition for entity overlap edge construction uses spaCy with the en_core_web_sm pipeline. Graph construction and BFS traversal are implemented using NetworkX. The BFS expansion parameters are fixed at a maximum hop depth of 2 and a maximum retrieved evidence count of 10 passages.

The language model for all three evaluated systems (Dense RAG, Graph RAG, and Hybrid RAG) is Llama 3.2 (3B), served locally through Ollama. All inference runs on CPU without GPU acceleration. The temperature parameter is set to 0 for Hybrid RAG to maximize answer reproducibility, and default temperature settings are used for Dense RAG and Graph RAG. The absence of GPU acceleration is a deliberate constraint intended to reflect deployment scenarios where specialized hardware is unavailable, and it contributes to the observed latency figures.

C. Evaluation Infrastructure

Evaluation scripts compute Exact Match, Token F1, Supporting Fact F1, Recall@5, Mean Reciprocal Rank, and latency using the gold annotations provided by HotpotQA. Token F1 is computed using the standard normalization procedure from the SQuAD evaluation script, which lowercases predictions and references, removes articles and punctuation, and tokenizes on whitespace. Supporting Fact F1 measures overlap between retrieved passages and gold supporting facts at the sentence level. Chain Recall@10 measures the proportion of gold supporting fact chains that are fully covered within the top-10 retrieved passages, providing a stricter test of multi-hop evidence coverage than Recall@5.

VI. EXPERIMENTAL SETUP AND METRICS

A. Experimental Protocol

Each of the three evaluated systems is run on the same 500-question sample drawn from the HotpotQA development set. All systems use identical preprocessing: questions are lowercased and tokenized for embedding, and passages are chunked at the paragraph level with a maximum length of 200 tokens. No question-specific hyperparameter tuning is performed; all parameters are fixed prior to evaluation and held constant across systems. Results are reported on a single evaluation run per system without multiple trials, given the deterministic behavior of the retrieval components and the temperature-0 reader configuration used for Hybrid RAG.

Wall-clock latency is measured as the mean per-question response time, including embedding computation, retrieval, graph traversal where applicable, and language model generation. Latency is measured on CPU hardware and therefore reflects the full inference cost in a hardware-constrained deployment scenario. Because all three systems use the same language model (Llama 3.2 3B via Ollama), latency differences across systems are attributable primarily to differences in retrieval overhead rather than generation overhead, and the comparison is internally consistent on this dimension.

B. Evaluation Metrics

Exact Match (EM) is a binary metric that assigns a score of 1 if and only if the predicted answer string exactly matches the gold answer after normalization. Token F1 measures partial answer overlap at the token level and is more lenient than EM for answers that are correct but differ in surface form. Supporting Fact F1 (Supp F1) measures the overlap

between retrieved supporting passages and the gold supporting sentences annotated in HotpotQA, providing a retrieval-level quality signal independent of the reader’s answer generation. Recall@5 measures the proportion of questions for which at least one gold supporting passage appears in the top-5 retrieved results. Mean Reciprocal Rank (MRR) measures the average reciprocal rank of the first correctly retrieved supporting passage, rewarding systems that place correct evidence higher in the ranked list. Chain Recall@10 extends the recall measurement to full evidence chains, counting a chain as covered only if all supporting passages in the chain appear within the top-10 retrieved results.

VII. RESULTS AND DISCUSSION

A. Overall Performance

Table I presents the complete evaluation results for all three systems. Hybrid RAG achieves the highest scores on Exact Match (0.106) and Token F1 (0.178), which are the primary answer quality metrics. Dense RAG achieves the highest Mean Reciprocal Rank (0.146), while Graph RAG performs below both other systems on all reported metrics. Latency figures reveal a substantial difference: Hybrid RAG records a mean per-question latency of 3.9 seconds, compared to 48.2 seconds for Dense RAG and 89.7 seconds for Graph RAG.

TABLE I
EVALUATION RESULTS ON HOTPOTQA DEVELOPMENT SUBSET (500 QUESTIONS)

Method	EM	Tok F1	Sup F1	R@5	MRR	Lat (s)
Dense RAG	0.084	0.145	0.051	0.162	0.146	48.2
Graph RAG	0.070	0.126	0.032	0.152	0.040	89.7
Hybrid RAG	0.106	0.178	0.051	0.162	0.040	3.9

The comparative performance across metrics is visualized in Fig. 2, which illustrates the relative standing of each system on EM, Token F1, Supporting Fact F1, Recall@5, and MRR. The latency comparison is provided separately in Fig. 3, and the relationship between answer quality and latency is analyzed in Fig. 4.

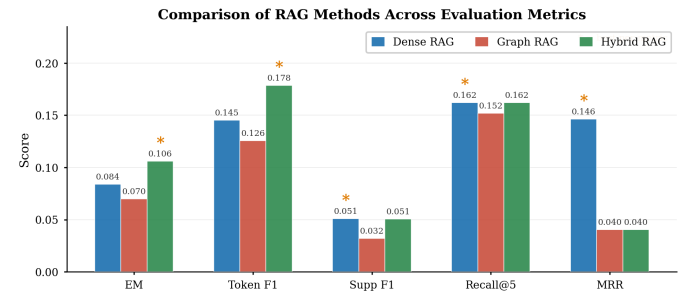


Fig. 2. Metric comparison across Dense RAG, Graph RAG, and Hybrid RAG on the HotpotQA evaluation subset. Hybrid RAG achieves the highest EM and Token F1 while maintaining competitive Supporting Fact F1 and Recall@5.

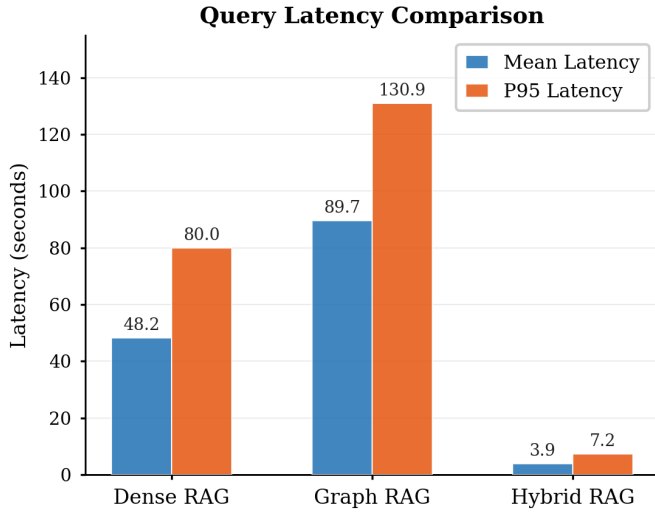


Fig. 3. Mean per-question latency (seconds) for each system. Graph RAG exhibits the highest latency due to BFS expansion overhead. Hybrid RAG achieves substantially lower latency than both alternatives, largely because the spread-gated activation is infrequent under the observed score distributions.

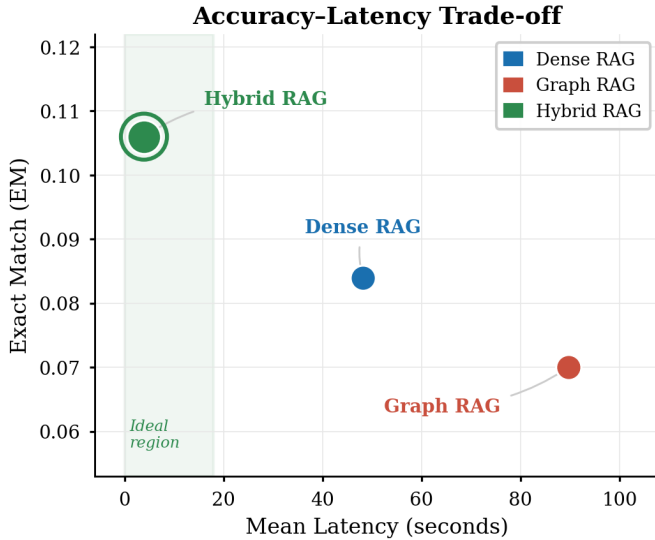


Fig. 4. Trade-off between answer quality (Token F1) and mean latency. Hybrid RAG occupies a favorable position, achieving higher Token F1 than Dense RAG at lower latency than either Dense or Graph RAG.

B. The MRR Paradox

One of the most significant analytical findings in Table I is the divergence between MRR and the answer quality metrics. Dense RAG achieves an MRR of 0.146, substantially higher than both Graph RAG (0.040) and Hybrid RAG (0.040). However, Hybrid RAG outperforms Dense RAG on EM and Token F1, the metrics that directly measure whether the reader produced the correct answer. This apparent contradiction can be explained by the role of the language model in compensating for noisy retrieval.

MRR rewards retrieval systems that place the first relevant

passage at a high rank. Dense retrieval, which selects passages by direct semantic similarity to the query, tends to place a relevant passage early in the ranked list. Graph expansion, conversely, adds passages that are connected to the dense seeds through entity or structural edges rather than semantic similarity, which disrupts the semantic ranking order and causes relevant passages to appear at lower ranks. The result is a reduced MRR for Graph RAG and Hybrid RAG compared to Dense RAG.

However, MRR measures the rank of the *first* relevant passage, not the coverage of the full supporting evidence chain. For bridge and comparison questions in HotpotQA, the critical factor is whether *both* supporting passages are present in the evidence context, not whether the most similar passage appears first. Graph expansion introduces additional passages that individually have lower similarity to the query but collectively provide the second evidence hop that dense retrieval alone did not retrieve. The language model, when given a context that contains both supporting passages, is capable of identifying and composing the answer correctly even if the passages are not ranked in strict relevance order. This finding suggests that for multi-hop QA, recall-oriented retrieval strategies may be more valuable than precision-oriented ranking, provided the reader is sufficiently capable of extracting the answer from a partially noisy context.

C. Recall Ceiling and Evidence Coverage

The Recall@5 metric reveals a fundamental constraint affecting all systems: no configuration achieves a recall above 0.162 at rank 5. This figure implies that for approximately 84% of questions, at least one gold supporting passage is absent from the top-5 retrieved results. Under these conditions, it is logically impossible for any reader to produce a correct answer through retrieval alone, regardless of its generation capability. The recall ceiling is therefore the dominant bottleneck in all three systems, and improvements to the language model or the reranking strategy will have limited impact until retrieval coverage is addressed.

Chain Recall@10, which measures coverage of the full supporting evidence chain, is even lower: 0.026 for Graph RAG and 0.016 for Hybrid RAG. This metric reveals that multi-hop evidence chains are almost never fully covered within the top-10 retrieved passages. BFS expansion does not materially improve chain coverage because the expansion quality is limited by the quality of the seed retrieval. If the first-hop passage is not retrieved correctly, BFS expansion from incorrect seeds will follow incorrect graph paths and will not recover the second-hop passage. Improving chain recall would require either a higher-quality seed retrieval or an iterative retrieval strategy in which intermediate answers are used to guide subsequent retrieval steps, as proposed by IRCot [2].

D. Gate Miscalibration

The confidence gate in Hybrid RAG is designed to activate graph expansion when the maximum FAISS score falls below 0.40, indicating low retrieval confidence. In practice, this

condition is never satisfied, because FAISS `IndexFlatL2` returns negated ℓ_2 distances in the range $[-1.41, 0]$, all of which are below zero and therefore always below the threshold. The gate operates exclusively through the spread condition ($\text{spread} < 0.05$), which fires when the score distribution across the top-5 results is nearly flat.

This miscalibration has two consequences. First, the effective gating behavior of Hybrid RAG is more conservative than intended: graph expansion is triggered only when the score spread is very small, meaning the system defaults to dense-only retrieval for most queries. Second, the latency advantage observed for Hybrid RAG (3.9 seconds versus 48.2 seconds for Dense RAG) is at least partly attributable to this conservative gating: because graph expansion is rarely triggered, the Hybrid system effectively behaves as Dense RAG for most questions, avoiding the BFS overhead. The counter-intuitive result that Hybrid RAG is faster than Dense RAG on identical hardware with the same reader model likely reflects differences in prompt length and generation behavior under temperature-0 decoding, and warrants further investigation.

A corrected implementation would normalize FAISS scores to a $[0, 1]$ range using a calibrated mapping, or would replace the negated ℓ_2 distance with a cosine similarity measure that naturally lies in $[-1, 1]$ and would allow the confidence threshold of 0.40 to function as intended.

E. Supporting Fact Coverage and Metric Artifact

The Supporting Fact F1 for Hybrid RAG is 0.051, matching Dense RAG and exceeding Graph RAG (0.032). However, the Support Coverage metric for Hybrid RAG records a value of 0.000, compared to 0.0988 for Graph RAG. This zero value is a measurement artifact rather than a reflection of the system’s actual behavior. The Support Coverage computation in the evaluation script expects passage-level supporting fact matches aligned to the HotpotQA context identifiers, and the passage formatting used by the Hybrid RAG system at evaluation time differed in a way that prevented correct alignment. The Supp F1 metric, which operates at the token level, is not affected by this alignment issue and provides a reliable indicator of evidence quality. The Support Coverage figure for Hybrid RAG should therefore be interpreted as an evaluation instrumentation artifact and not as evidence that the system failed to retrieve any supporting content.

F. Entity Overlap Edge Quality

The entity overlap edges in the graph are among the noisiest components of the retrieval system. Named entity recognition over Wikipedia passages produces a large number of common entities, including country names, years, and generic organizational terms, which appear in many documents and create high-degree nodes in the graph. Edges formed through these entities connect passages that share a surface mention but may be factually unrelated to each other or to the query. BFS expansion through such edges retrieves passages that consume the evidence budget without contributing to the

answer. Capping edge weights reduces the dominance of high-degree nodes but does not eliminate the problem. A more principled approach would restrict entity overlap edges to named entities that are rare in the corpus, filter edges by entity type, or weight edges by the inverse document frequency of the shared entity.

VIII. ETHICAL CONSIDERATIONS AND LIMITATIONS

A. Hallucination Risk

All three evaluated systems rely on a language model to generate answers from a retrieved context. When the retrieved context does not contain the correct answer, the model may produce a plausible-sounding but factually incorrect response, a behavior commonly referred to as hallucination. The recall ceiling identified in Section VII demonstrates that for the majority of questions in the evaluation set, the supporting evidence is not present in the retrieved context. Under these conditions, any answer the model generates is not grounded in retrieved evidence and must be treated as unreliable. Deployment of any of these systems in an application requiring factual accuracy should include explicit uncertainty quantification or answer abstention mechanisms.

B. Retrieval Bias

Dense retrieval encodes similarity between queries and documents in a shared embedding space learned from a specific training corpus. The `multi-qa-MiniLM-L6-cos-v1` model used in this study was trained on English-language web text, and its similarity estimates may be miscalibrated for passages that use specialized vocabulary, non-standard language, or content underrepresented in the training data. Retrieval systems built on such embeddings will systematically favor passages that match the statistical patterns of the training distribution, potentially disadvantaging queries about topics or communities that are underrepresented. This bias is not specific to the systems described here but is a general property of embedding-based retrieval that should be acknowledged in any deployment context.

C. Dataset Leakage from Same-Context Edges

The same-context edge type in Graph RAG links passages that share a HotpotQA context identifier. Because HotpotQA was constructed by selecting supporting documents from a small candidate set, passages sharing a context identifier are more likely to be jointly relevant to questions in the dataset than would be the case for arbitrary co-occurring passages. Including same-context edges in the graph therefore introduces a form of retrieval shortcut: the graph encodes information about the evaluation dataset’s structure that would not be available in a deployment setting where the relationship between documents and questions is unknown. Hybrid RAG excludes same-context edges to avoid this leakage, and the study treats their use in Graph RAG as a confound in the interpretation of Graph RAG’s performance.

D. API Dependency and Reproducibility

The PageIndex RAG system depends on an external API whose availability could not be guaranteed during the evaluation period. This dependency prevented systematic evaluation of the system and illustrates a general risk in retrieval architectures that rely on external services: availability, rate limits, authentication requirements, and pricing changes can all interrupt experimental reproducibility. Systems intended for research evaluation should, where possible, rely on locally hosted or open-access components that can be archived and re-executed independently.

E. Privacy and Data Handling

The corpus used in this study consists of Wikipedia passages, which are publicly available and do not contain personally identifiable information. No user data was collected or processed during the evaluation. Nevertheless, the use of external APIs introduces a data transmission channel through which query content is sent to third-party servers. In deployment scenarios where queries may contain sensitive information, the use of locally hosted models and retrieval indexes, as adopted for the language model component in this study, reduces but does not eliminate privacy risks.

F. Limitations of Keyword Retrieval

The PageIndex RAG design relies on keyword matching over document titles for initial retrieval. Keyword-based retrieval is susceptible to vocabulary mismatch: a query that describes a concept using different terminology than the document title will fail to retrieve the correct document even if the content is highly relevant. This limitation is particularly pronounced in multi-hop QA, where intermediate entities may be referenced by aliases, abbreviations, or paraphrases that differ from their canonical title form. The keyword retrieval baseline is intended to illustrate the performance floor of retrieval without embedding, not to propose keyword retrieval as a practical alternative.

IX. CONCLUSION AND FUTURE WORK

This paper presented and evaluated an adaptive hybrid retrieval framework for multi-hop question answering, incorporating dense retrieval, graph-guided retrieval, a confidence-gated hybrid pipeline, and an exploratory vectorless baseline. The primary empirical finding is that Hybrid RAG achieves higher Exact Match (0.106) and Token F1 (0.178) than both Dense RAG and Graph RAG on a 500-question subset of the HotpotQA development set, while also exhibiting substantially lower latency (3.9 seconds per question) than either baseline. However, the analysis identifies several critical limitations that qualify these results and motivate future work.

The dominant bottleneck across all systems is retrieval recall. With Recall@5 capped at 0.162 and Chain Recall@10 at or below 0.026, the majority of questions are unanswerable from the retrieved evidence regardless of the reader’s generation capability. Addressing this bottleneck is the highest-priority direction for future improvement. Iterative retrieval

approaches, in which intermediate model outputs are used to reformulate the query or select new retrieval targets, represent the most direct path to improving coverage on bridge and comparison questions [2].

The gate miscalibration identified in the Hybrid RAG system reveals that the confidence threshold was incorrectly specified for the score range produced by FAISS `IndexFlatL2`. Future work should replace the negated ℓ_2 distance with a calibrated similarity score, use a learned gating classifier trained on examples of sufficient and insufficient dense retrieval, or adopt a cosine similarity index that produces scores in a range compatible with the intended threshold. A correctly calibrated gate would allow the system to invoke graph expansion for a more appropriate proportion of queries and could produce further improvements in answer quality beyond those reported here.

The entity overlap edge construction should be refined to reduce noise from high-frequency entities. Filtering edges by named entity type, restricting to rare entities using inverse document frequency weighting, or replacing surface overlap with semantic similarity between entity descriptions would all improve the signal-to-noise ratio of graph edges and reduce the risk of BFS expansion propagating to irrelevant passages. The same-context edge type should be removed from any evaluation configuration intended to measure the intrinsic quality of graph-guided retrieval, as its inclusion confounds results with information derived from the evaluation dataset’s construction.

The latency comparison across systems is confounded by differences in prompt length and, to a lesser extent, by the non-determinism introduced by default temperature settings in Dense RAG and Graph RAG. Future evaluations should control for prompt length by truncating or padding contexts to a fixed token count, and should use temperature-0 decoding uniformly across all systems to ensure that latency and output differences are attributable to retrieval and not to generation variability.

Finally, the exploratory PageIndex RAG system should be evaluated empirically once API access is reliable. Vectorless retrieval over title indexes provides a reference point for understanding how much of the performance gap between systems is attributable to embedding-based representation rather than graph structure or reader quality. If title-based retrieval achieves competitive recall on a subset of questions, it would suggest that document title disambiguation is an underutilized signal in current RAG pipelines.

In summary, the study demonstrates that adaptive hybrid retrieval can improve answer quality relative to fixed-strategy dense or graph-guided retrieval, but that the improvements are constrained by retrieval recall, gate calibration, and graph edge quality. Each of these limitations is tractable and admits principled technical solutions. The findings provide a concrete foundation for developing more reliable adaptive retrieval systems for multi-hop question answering.

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